

H: Only1s0s

Source file name: `only1s0s.c`, `only1s0s.cpp`, `only1s0s.java`, or `only1s0s.py`

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Alana is a very curious number theory researcher. Last week, she discovered a --perhaps-- useless property about positive integers: for every such number N , there exists an integer M , a multiple of N , that consists only of the digits 1 and 0 in decimal notation. This number M is called an `only1s0s` number.

For instance:

- If $N = 4$, then $M = 10000$ and $\frac{M}{N} = 2500$.
- If $N = 14$, then $M = 10010$ and $\frac{M}{N} = 715$.

Alana has proven that there are many possible solutions for any N . However, she is interested only in the smallest possible M for a given N . Moreover, she wants to determine the value D such that $M = N \cdot D$, where M is the minimum `only1s0s` number.

Can you help her?

Input

The input consists of several test cases. Each test case is defined by one line containing a positive integer N ($0 < N < 10^5$). The input ends with a line containing 0.

The input must be read from standard input.

Output

For each test case, output a single line containing the positive integer D , such that $N \cdot D$ is the smallest `only1s0s` number that is a multiple of N .

The output must be written to standard output.

Sample Input	Sample Output
4	25
14	715
13	77
0	